VMCraft Engine Deep Dive

Pure Python Conversion Without libguestfs

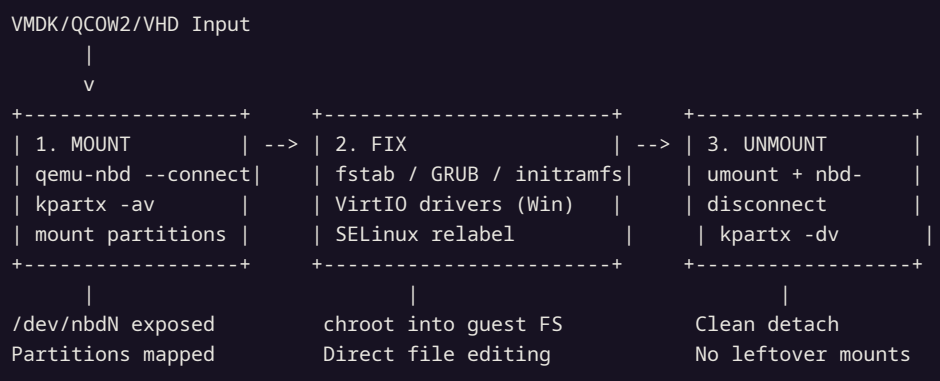
VMCraft is hyper2kvm's default conversion backend. It uses qemu-nbd for block device mounting, direct filesystem access for offline fixes, and chroot for initramfs regeneration. No supermin appliance, no 500MB dependency chain, no appliance boot delay. Just Python + qemu-nbd.

qemu-nbd | chroot | fstab | GRUB | VirtIO | Windows Registry | ~5MB footprint

The Mount → Fix → Unmount Pipeline

VMCraft's three-phase approach to offline VM repair.

Pipeline Overview



Phase 1: Mount — qemu-nbd Block Device Exposure

qemu-nbd Connection

- `modprobe nbd max_part=16` — load NBD kernel module
- `qemu-nbd --connect=/dev/nbd0 --format=vmdk disk.vmdk`
- Supports VMDK (monolithic/split), QCOW2, VHD, VHDX, raw
- Read-write access to guest disk as a local block device
- VMDK descriptor files auto-resolved (sparse/flat extents)
- No image conversion needed — operates on source format directly

Partition Discovery

- `kpartx -av /dev/nbd0` — creates /dev/mapper/nbd0p1, nbd0p2...
- Handles MBR and GPT partition tables automatically
- LVM detection: `vgscan` + `vgchange -ay` to activate VGs
- Root partition identified by filesystem label, UUID, or content heuristics
- EFI System Partition (ESP) detected for UEFI boot repair
- Swap partitions identified and skipped during mount

Phase 2: Fix — Offline Guest Filesystem Repair

fstab Repair with blkid

- VMware device names (sda) become VirtIO names (vda)
- `blkid /dev/mapper/nbd0p*` reads UUIDs from actual partitions
- Rewrites /etc/fstab: `/dev/sda1` → `UUID=xxxx-xxxx`

GRUB Configuration

- Detects GRUB2 (grub.cfg) and legacy GRUB (menu.lst)
- Rewrites root= kernel parameter: `root=/dev/sda2` → `root=UUID=...`
- Adds VirtIO modules to GRUB_PRELOAD_MODULES if needed

- Handles ext4, xfs, btrfs, swap entries
- Preserves mount options, dump/fsck fields
- Backs up original fstab before modification

- Regenerates grub.cfg via chroot: `grub2-mkconfig -o /boot/grub2/grub.cfg`
- UEFI: updates EFI/fedora/grub.cfg or EFI/redhat/grub.cfg
- Handles BLS (Boot Loader Spec) entries on Fedora/RHEL 8+

Initramfs Rebuild via chroot

```
# VMCraft chroot sequence for initramfs regeneration
mount --bind /dev  /mnt/guest/dev
mount --bind /proc /mnt/guest/proc
mount --bind /sys  /mnt/guest/sys

# Detect distro and kernel version
KVER=$(ls /mnt/guest/lib/modules/ | sort -V | tail -1)

# RHEL/Fedora: dracut
chroot /mnt/guest dracut --force --add-drivers "virtio_blk virtio_net virtio_pci virtio_scsi" \
    /boot/initramfs-${KVER}.img ${KVER}

# Debian/Ubuntu: update-initramfs
chroot /mnt/guest update-initramfs -u -k ${KVER}

# Cleanup bind mounts
umount /mnt/guest/{sys,proc,dev}
```

Windows Migration: Registry & VirtIO

VMCraft handles Windows VMs without requiring a running Windows instance.

Windows Registry Hive Loading

Offline Registry Editing

- Mount NTFS partition: `ntfs-3g /dev/mapper/nbd0p2 /mnt/guest`
- Load SYSTEM hive: `hivexregedit --merge` or Python hivex bindings
- Navigate to `HKLM\SYSTEM\CurrentControlSet\Services`
- Set VirtIO driver services to Start=0 (boot start)
- Services: viostor, vioscsi, vionet, balloon, vioserial
- Remove VMware Tools services: vmtoolsd, vmhgfs, VMCI

VirtIO Driver Extraction from ISO

- Downloads or uses local virtio-win.iso (Red Hat signed drivers)
- Mounts ISO: `mount -o loop virtio-win.iso /mnt/virtio`
- Detects Windows version from registry: Win10, Win11, 2019, 2022
- Copies matching drivers: `/mnt/virtio/amd64/w10/viostor.sys`
- Installs .inf files into driver store: `Windows/INF/oem*.inf`
- Creates CriticalDeviceDatabase entries for boot-time loading

Windows VirtIO Registry Entries (Created by VMCraft)

```
; VirtIO SCSI boot driver — prevents BSOD 0x7B (INACCESSIBLE_BOOT_DEVICE)
[HKLM\SYSTEM\CurrentControlSet\Services\viostor]
"Type"=dword:00000001
"Start"=dword:00000000          ; 0 = Boot start (loaded by bootloader)
"ErrorControl"=dword:00000001
"Group"="SCSI miniport"
"ImagePath"="system32\\drivers\\viostor.sys"

; VirtIO Network — prevents "No network adapters" after migration
[HKLM\SYSTEM\CurrentControlSet\Services\netkvm]
"Type"=dword:00000001
"Start"=dword:00000003          ; 3 = Manual (PnP will start it)
"ImagePath"="system32\\drivers\\netkvm.sys"

; Critical Device Database — maps PCI VEN/DEV to driver
[HKLM\SYSTEM\CurrentControlSet\Control\CriticalDeviceDatabase\pci#ven_1af4&dev_1001]
"Service"="viostor"
"ClassGUID"="{4D36E97B-E325-11CE-BFC1-08002BE10318}"
```

VMCraft vs libguestfs: Architecture Comparison

Aspect	VMCraft (hyper2kvm default)	libguestfs

How it works	qemu-nbd exposes disk as /dev/nbdN, mount directly on host	Boots a supermin appliance (mini Linux VM), accesses disk inside appliance
Disk footprint	**~5 MB (qemu-nbd binary)**	**~500 MB (supermin appliance + kernel + initrd)**
Startup time	**Instant (~100ms for nbd connect)**	**15-45 seconds (appliance boot)**
Dependencies	**qemu-utils, kpartx, Python 3**	**libguestfs, supermin, hivex, libvirt, kernel, febootstrap**
Air-gap install	**3 RPMs or DEBs**	**30+ RPMs, supermin rebuild may need network**
Root required	Yes (nbd, mount, chroot)	No (runs in appliance VM)
SELinux compat	**Works with enforcing (relabels guest)**	Appliance may conflict with host SELinux
Windows support	**Full VirtIO injection + registry editing**	**virt-v2v handles Windows (different tool)**
Performance	**Native I/O speed (direct mount)**	Slightly slower (virtio inside appliance)
Debugging	**Standard Linux tools (ls, cat, strace)**	Must debug inside appliance (guestfish shell)

Complete Conversion Flow

Step-by-step walkthrough of a Linux VM conversion using VMCraft.

Example: Converting a RHEL 8 VMDK to KVM

```
$ h2kvmctl convert --source disk.vmdk --output /var/lib/libvirt/images/ --format qcow2 \
    --regen-initramfs --emit-domain-xml

[1/8] Validating source: disk.vmdk (VMDK, 40GB, thin provisioned)
[2/8] Converting VMDK -> QCOW2: qemu-img convert -p -f vmdk -O qcow2 -c disk.vmdk output.qcow2
      Progress: [========================================] 100%  38.2 GB  @ 245 MB/s
[3/8] Connecting qemu-nbd: /dev/nbd0 -> output.qcow2
[4/8] Discovering partitions: nbd0p1 (EFI, 600MB), nbd0p2 (xfs, 39.4GB)
[5/8] Mounting root: /dev/mapper/nbd0p2 -> /tmp/hyper2kvm-abc123/mnt
      Mounting EFI:  /dev/mapper/nbd0p1 -> /tmp/hyper2kvm-abc123/mnt/boot/efi
[6/8] Applying fixes:
      - fstab: /dev/sda2 -> UUID=a1b2c3d4-e5f6-7890-abcd-ef1234567890
      - fstab: /dev/sda1 -> UUID=ABCD-1234 (vfat)
      - GRUB: root=/dev/sda2 -> root=UUID=a1b2c3d4-e5f6-7890-abcd-ef1234567890
      - initramfs: adding virtio_blk virtio_net virtio_pci virtio_scsi
      - dracut: regenerating /boot/initramfs-4.18.0-513.el8.x86_64.img
      - SELinux: touching /.autorelabel
[7/8] Unmounting: nbd0p2, nbd0p1, nbd-disconnect
[8/8] Generating libvirt XML: output.xml

Conversion complete: output.qcow2 (14.2 GB on disk, 40 GB virtual)
```

Code Architecture: Key Python Modules

vmcraft/mount.py

- `NbdManager` — context manager for qemu-nbd lifecycle
- Finds next free /dev/nbdN (checks /sys/block/nbdN/size)
- `connect(path, fmt)` — qemu-nbd --connect with retry
- `disconnect()` — qemu-nbd --disconnect with verification
- Timeout: waits up to 10s for /dev/nbdNp* to appear
- Cleanup: registered with atexit for crash safety

vmcraft/partitions.py

- `discover_partitions(nbd_dev)` — kpartx + blkid probe
- Returns list of Partition objects with UUID, fstype, label, size
- `find_root(partitions)` — heuristic: largest ext4/xfs/btrfs
- `find_efi(partitions)` — vfat partition with EFI flag
- LVM: scans for LV paths under /dev/mapper/vgname-lvname
- Handles multi-disk VMs by iterating nbd0, nbd1, ...

vmcraft/fixer.py

vmcraft/convert.py

- `fix_fstab(root_mount)` — UUID-based rewrite
- `fix_grub(root_mount)` — kernel cmdline + grub.cfg regen
- `rebuild_initramfs(root_mount, kver)` — chroot dracut/initramfs
- `inject_virtio_windows(root_mount)` — registry + driver copy
- `set_selinux_relabel(root_mount)` — touch /.autorelabel
- `remove_vmware_tools(root_mount)` — cleanup VMware artifacts

- `qemu_img_convert(src, dst, fmt, compress)` — format conversion
- Progress parsing: reads qemu-img stdout for percentage
- Supports: vmdk, qcow2, vhd, vhdx, raw output formats
- Compression: -c flag for qcow2 (zlib, ~40% size reduction)
- Checksum: SHA256 of output image for verification
- Thin provisioning: qcow2 inherently thin (sparse allocation)

Error Handling & Recovery

Crash Safety: VMCraft registers cleanup handlers with Python's `atexit` module. If the process crashes or is interrupted (Ctrl+C), the cleanup sequence runs: `umount -l` (lazy unmount), `kpartx -dv` (remove mappings), `qemu-nbd --disconnect` (release NBD). LVM volume groups are deactivated with `vgchange -an`. Temporary directories under `/tmp/hyper2kvm-*` are removed. This prevents stale mounts and NBD device leaks.

Advanced VMCraft Features

Beyond basic conversion: OVA extraction, multi-disk VMs, and performance tuning.

OVA/OVF Processing Pipeline

OVA Extraction

- OVA = tar archive containing .ovf + .vmdk + .mf (manifest)
- VMCraft streams extraction: reads tar members without full unpack
- Validates SHA256 checksums from .mf manifest file
- Parses .ovf XML for VM config: CPU, RAM, disk controllers
- Maps OVF hardware to libvirt XML: `VirtualSCSI → virtio-scsi`
- Handles split VMDKs: reassembles descriptor + extent files

Multi-Disk VM Handling

- Each VMDK gets its own /dev/nbdN (nbd0, nbd1, nbd2...)
- Root disk identified from OVF boot order or partition content
- Data disks: converted and attached to libvirt XML as vdb, vdc...
- Cross-disk LVM: VGs spanning multiple PVs detected and activated
- Independent fstab entries for each data disk
- Max 16 NBD devices (configurable via nbd module parameter)

Performance Characteristics

Operation	Time (40GB disk)	Bottleneck	Notes
qemu-img convert (VMDK→QCOW2)	2-3 min	Disk I/O	NVMe source: ~250 MB/s, SATA: ~150 MB/s
qemu-nbd connect	<1 sec	Kernel	NBD module load + device setup
kpartx + mount	<2 sec	Kernel	Partition table scan + filesystem mount
fstab + GRUB fix	<1 sec	CPU	Text parsing + blkid lookup
initramfs rebuild (dracut)	15-45 sec	CPU/Disk	Kernel module collection + cpio archive
Unmount + disconnect	<2 sec	Kernel	Sync + unmount + NBD release
Total (Linux VM)	**3-5 min**	Disk I/O	Dominated by qemu-img convert
Total (Windows VM)	**4-7 min**	Disk I/O	+1-2 min for VirtIO driver copy + registry

Supported Guest Operating Systems

Linux (Full Support)

Windows (Full Support)

- RHEL / CentOS / Rocky / Alma 7, 8, 9
- Fedora 36+
- Ubuntu 18.04, 20.04, 22.04, 24.04
- Debian 10, 11, 12
- SLES / openSUSE 15+
- VMware Photon OS 3, 4, 5

- Windows Server 2016, 2019, 2022, 2025
- Windows 10, 11 (Pro, Enterprise)
- VirtIO drivers: viostor, vioscsi, netkvm, balloon
- Automatic BSOD prevention (0x7B, 0x5C)
- VMware Tools removal
- Hyper-V enlightenments preserved

VMCraft: Simple, Fast, Reliable

No appliance boot. No 500MB dependency chain. No supermin.
Mount the disk, fix the files, unmount. Pure Python + qemu-nbd.
~5MB footprint. 3-5 minute conversions. Works in air-gapped environments.